Erfan Jaripour

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Education

Bachelor of Science in Psychology, Islamic Azad University, Tehran, Iran 2021–2025
GPA: 19.55/20 (equivalent approximately 3.9/4.0)

Thesis: Relationship Between Psychological Disorders and Defense Mechanisms in Adolescents
Supervisor: Dr. Hamid Reza Vatankhah

Selected Coursework: Descriptive Statistics; Inferential Statistics; Cognitive Psychology; Experimental Psychology; Psychometrics; Nurological Sciences; Research in Psychology

Additional Quantitative Preparation: Linear Algebra; Calculusc; Probability Teory; Scientific Programming with Python and R; Data Analysis and Viualisation

Research Interests

Computational Cognitive science; Working Memory; Learning and Adaptive Behavior; Decision-Making; Reinforcement Learning; Bayesian Cognitive Modeling.

Research Experience

Independent Computational Research Projects 2025–Present

Stroop Interference: A Behavioral and Computational Analysis June 2025

- Analyzed trial-level Stroop task data from 81 participants using Python.
- Cleaned and validated behavioral datasets, including participant–session structure resolution and quality-control procedures.
- Quantified Stroop interference through descriptive statistics, paired-samples inference, effect-size estimation, and robustness analyses.
- Examined individual differences in interference magnitude across participants.
- Implemented a simple computational simulation to reproduce observed condition-level behavioral patterns.
- Produced a fully reproducible research workflow with documented analyses, visualizations, and manuscript preparation.

Repository: github.com/erfanjaripour/stroop-interference

Computational Cognitive Science Training Mar 2025

- Structured self-directed training in Python-based data analysis for cognitive science and behavioral research.
- Developed practical skills in NumPy, Pandas, Matplotlib/Seaborn, and inferential statistics.
- Completed mini-projects after each module, including data simulation, cleaning, visualization, and hypothesis testing.
- Built reproducible analysis workflows with clear notebook organization and documented code.
- Applied statistical methods such as descriptive analysis, regression, and basic model interpretation to research-oriented datasets.

Repository: github.com/erfanjaripour/computational-cognitive-science-training

Publications & Preprints

Jaripour, E. (2026). *Stroop Interference in Reaction Time and Accuracy: A Behavioral and Computational Analysis.* PsyArXiv preprint.

Technical Skills

PROGRAMMING: Python, R
SCIENTIFIC LIBRARIES: pandas, NumPy, scipy, matplotlib, seaborn, statsmodels
STATISTICAL METHODS: descriptive statistics, inferential statistics, hypothesis testing, ANOVA, regression, repeated-measures designs, effect-size estimation, confidence intervals, exploratory data analysis
RESEARCH WORKFLOW: Git, GitHub, Jupyter Notebooks, Markdown, LaTeX, reproducible research

Honors & Awards

Top-Ranked Psychology Student, Islamic Azad University	2021–2025
Tuition Waiver Scholarship, Islamic Azad University	2024
Outstanding Academic Performance Award, Islamic Azad University	2023

Languages

Persian (Native)
English (Professional Working Proficiency)

References

Available upon request.